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// CITATION: CHATGPT HAS ASSISTED IN THE DEBUG PROCESS AND CREATION OF THE FOR LOOP
// IN DEPTH CITATIONS WILL FOLLOW

// A list created for all the symbols used when making math calculations
var symbolList = ["+", "-", "*", "/", "=", "F", "C", "cm", "in"];
// A blank list to be later used to add the history of previous calculations
var calculation_history = [];

// When the user clicks the "Enter!" button, they will move to the screen to make calculations
// It will also hide all elements that are used when inputing to make calculations initially
onEvent("enter_button", "click", function( ) {
  setScreen("calculations_screen");
  hideElement("label1");
  hideElement("label2");
  hideElement("input1");
  hideElement("input2");
  hideElement("output1");
  hideElement("calculate_button");
  hideElement("didyouknow");
  hideElement("first_reading");
  hideElement("first_temp");
  hideElement("result_temp");
  hideElement("second_reading");
  hideElement("label5");
  hideElement("history_label");
});

// When the user clicks on any of the types of calculations in the dropdown menu
// The elements used to make the calculations will appear
onEvent("select_calc", "click", function( ) {
  operate();
});
// When the user clicks on the "Calculate!" button, an output (answer) will be outputted when they put the needed values
// in the input boxes that appear
// The answer that was produced will also be added to the calculation history
onEvent("calculate_button", "click", function( ) {
operate();
calc_hist();
});

// Based on the calculation type the user they choose on the dropdown
// The symbols of that calculation type are inputted and a specific formula is used
function operate(dropdown, num1, num2, reading1) {
dropdown = getText("select_calc");
num1 = getNumber("input1");
num2 = getNumber("input2");
reading1 = getNumber("first_temp");
var output;

// When they select "Add", the plus symbol and the equal symbol are inputted
// When the user inputs the neccessary values, it will add them
if (dropdown == "Add") {
  setText("label1", symbolList[0]);
  setText("label2", symbolList[4]);
  establish_basic();
  output = num1 + num2;
// When they select "Subtract", the minus symbol and the equal symbol are inputted
// When the user inputs the neccessary values, it will subtract them
  } else if ((dropdown == "Subtract")) {
  setText("label1", symbolList[1]);
  setText("label2", symbolList[4]);
  establish_basic();
  output = num1 - num2;
// When they select "Multiply", the times symbol and the equal symbol are inputted
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// When the user inputs the neccessary values, it will multiply them
  } else if ((dropdown == "Multiply")) {
  setText("label1", symbolList[2]);
  setText("label2", symbolList[4]);
  establish_basic();
  output = num1 * num2;
// When they select "Divide", the minus symbol and the equal symbol are inputted
// When the user inputs the neccessary values, it will subtract them
  } else if ((dropdown == "Divide")) {
  setText("label1", symbolList[3]);
  setText("label2", symbolList[4]);
  establish_basic();
  output = num1 / num2;
// When they select "Fahrenheit to Celsius", the Fahrenheit, Celsius, and equal symbols are inputted
// When the user inputs the neccessary values, it will convert Fahrenheit to Celsius
  } else if ((dropdown == "Fahrenheit to Celsius")) {
  setText("first_reading", symbolList[5]);
  setText("second_reading", symbolList[6]);
  setText("label5", symbolList[4]);
  establish_conv();
  output = (reading1 - 32) * 5/9;
// When they select "Celsius to Fahrenheit", the Fahrenheit, Celsius, and equal symbols are inputted
// When the user inputs the neccessary values, it will convert Celsius to Fahrenheit
  } else if ((dropdown == "Celsius to Fahrenheit")) {
  setText("first_reading", symbolList[6]);
  setText("second_reading", symbolList[5]);
  setText("label5", symbolList[4]);
  establish_conv();
  output = (reading1 * 9/5) + 32;
// When they select "Inches to Centimeters", the Inches, Centimeters, and equal symbols are inputted
// When the user inputs the neccessary values, it will convert Inches to Centimeters
  } else if ((dropdown == "Inches to Centimeters")) {
  setText("first_reading", symbolList[8]);
  setText("second_reading", symbolList[7]);
  setText("label5", symbolList[4]);
  establish_conv();
  output = (reading1 * 2.54);
// When they select "Centimeters to Inches", the Inches, Centimeters, and equal symbols are inputted
// When the user inputs the neccessary values, it will convert Centimeters to Inches
  } else if ((dropdown == "Centimeters to Inches")) {
  setText("first_reading", symbolList[7]);
  setText("second_reading", symbolList[8]);
  setText("label5", symbolList[4]);
  establish_conv();
  output = (reading1 / 2.54);
// When they select "Power", the equal symbol is inputted
// When the user inputs the neccessary values, it will take the value and multiply it by the same value
  } else if ((dropdown == "Power")) {
  setText("label5", symbolList[4]);
  establish_conv();
  hideElement("first_reading");
  hideElement("second_reading");
  setPosition("label5", 145, 130, 40, 30);
  output = reading1 * reading1;
  }
// They both set the answer whenever a calculation is made whether it is a normal math calculation or a conversion
setText("output1", output);
setNumber("result_temp", output);
}

// It will hide and show the necessary elements that are required when doing simple math like
// Adding, subtracting, multiplying, or dividing
function establish_basic() {
  showElement("didyouknow");
  showElement("input1");
  showElement("input2");
  showElement("label2");
  showElement("output1");
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  showElement("label1");
  showElement("input2");
  showElement("label2");
  showElement("output1");
  showElement("calculate_button");
  showElement("history_label");
  hideElement("first_temp");
  hideElement("result_temp");
  hideElement("first_reading");
  hideElement("label5");
  hideElement("second_reading");
}
// It will hide and show the neccesary elements that are required when doing conversions
// Such as temperatures and distance measurements
// It is also used when selecting "Power" which is just multiplying the same number twice
function establish_conv() {
  hideElement("input1");
  hideElement("input2");
  hideElement("label2");
  hideElement("output1");
  hideElement("label1");
  hideElement("input2");
  hideElement("label2");
  hideElement("output1");
  showElement("didyouknow");
  showElement("calculate_button");
  showElement("first_temp");
  showElement("result_temp");
  showElement("first_reading");
  showElement("label5");
  showElement("second_reading");
  showElement("history_label");
}

// When the user makes a calculation, the value will be added to a list that will
// Create a history of all the calculations the user makes

// CITATION: CHATGPT - LINE 178. The AI created the called variable to work and allowed the list to be in the string
// CITATION: CHATGPT - LINE 176. The AI assisted in bebugging an issue where the list would not print as a string.
// It created a variable to be able to make the list be printed out as a string.
function calc_hist() {
    var output_value = getNumber("output1");
appendItem(calculation_history, output_value);
    var historyList = " ";
    for (var i = 0; i < calculation_history.length; i++) {
        historyList = historyList + calculation_history[i] + ", ";
    }
    setText("didyouknow", historyList);
}
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